Critical Technocultural Discourse Analysis via Open WebUI: Architectural Configuration and Deployment on Reddit

The empirical study of digital platforms and internet phenomena requires analytical frameworks capable of unpacking the complex, deeply entrenched intersections between technological architecture and cultural practice. Historically, discourse analysis within digital environments has defaulted to universalist assumptions. When analysts make broad claims regarding "what people do online," the implicit demographic center of these claims is generally the white, Western, upper-middle-class user from economically dominant nations.[1] Studies that diverge from this normative population are forced to mark themselves as specific deviations, whereas studies maintaining this hegemonic hue are presented as universal.[1] This "whitewashing of generality" not only obscures and reifies racial, class, and national hegemonies but fundamentally entrenches these biases into entire theoretical systems, resulting in research that is both theoretically misleading and politically regressive.[1]

To counter these normative assumptions and establish a rigorous, equitable mechanism for digital inquiry, Critical Technocultural Discourse Analysis (CTDA) provides a vital theoretical intervention. Developed by André Brock, CTDA is a multimodal analytic technique explicitly designed to investigate Internet phenomena, digital artifacts, and online culture by prioritizing the epistemological standpoints of marginalized and underrepresented groups.[2]

Operationalizing CTDA at scale, however, presents significant computational challenges. It requires the ingestion of massive volumes of unstructured platform data, the contextual isolation of specific technocultural enclaves, and the application of highly nuanced qualitative evaluation.

The deployment of Large Language Models (LLMs) offers a potent mechanism to process this qualitative data at scale. Yet, out-of-the-box LLMs are prone to hallucinating analytical frameworks or defaulting to the very normative, deficit-based analytical models that CTDA seeks to dismantle.[2] Therefore, executing a true, mathematically and theoretically sound CTDA requires a tightly controlled, self-hosted computational environment where the model's instructions, knowledge base, and data extraction tools are explicitly engineered to adhere to Brock's framework.

Open WebUI serves as an optimal temporary interface and orchestration layer for this endeavor. As an extensible, feature-rich, and self-hosted AI platform designed to operate entirely offline, it allows researchers to orchestrate Ollama or OpenAI-compatible models within a secure, localized environment, effectively neutralizing the risk of data leakage or external algorithmic interference.[5] By configuring a specialized Modelfile, engineering explicit "Collection Skills," reducing the system prompt recommendations for knowledge collection to a

"one scale" matrix, and injecting custom Python code for Reddit data extraction via the Python Reddit API Wrapper (PRAW) [6], researchers can construct an autonomous, localized agent capable of executing rigorous CTDA. This report details the comprehensive theoretical grounding, structural deployment, and code-level implementation required to build this techno-cultural analysis instrument.

Theoretical Foundations: Critical Technocultural Discourse Analysis

Critical Technocultural Discourse Analysis demands that technologies be understood not as neutral, mathematical tools residing outside of society, but as culturally embedded assemblages of artifacts, practices, and beliefs.[1] It represents an aggressive intervention into normative technology analyses by insisting that culture and technology must be simultaneously interrogated as deeply intertwined, mutually constitutive concepts.[2] The framework incorporates critical theory—specifically critical race theory, feminism, and queer theory—to actively avoid and dismantle deficit-based models that historically frame minority technology use as pathological, deviant, or perpetually lagging behind a white, patriarchal norm.[1]

A core component of CTDA is its reliance on the concept of the "libidinal economy." Originating with Freud and further developed by philosophers such as Jean-François Lyotard, Fred Moten, and Frank B. Wilderson III, the libidinal-economic approach emphasizes the role of emotional and psychological intensities in driving societal structures, including anti-Blackness, rather than relying on strictly rationalist models of human behavior.[3] CTDA applies this to digital spaces to understand the affective artifacts charged with emotional labor, cultural memory, and strategic resistance.[8]

To operationalize this intervention computationally, CTDA deconstructs digital phenomena across four specific, interconnected dimensions. A machine learning model designed to perform CTDA must be explicitly trained to recognize, isolate, and categorize unstructured platform data into these exact vectors.[1]

CTDA Dimension	Conceptual Definition and Scope	Computational Application to Digital Analysis
Form	The material and architectural nature of the technology. This encompasses the hardware, the code, the user interface, database structures, and the physical/digital constraints of the platform itself.	Examining how algorithmic sorting, character limits, API rate limits, or voting mechanisms dictate visibility and shape user interaction parameters.[1]
Function	The practices and interactions that occur between the technological artifacts, cultural conventions, and personal	Analyzing specific, community-bound user behaviors, such as the deployment of memes,

	experiences. It assumes technology use is contextually contingent.	hashtags, linguistic styles, or specific comment-threading behaviors within a bounded enclave.[1]
Belief	The cultural ideologies embedded within the technology's design and the deeply held beliefs surrounding its use, particularly those reflecting or challenging systemic hegemony and power structures.	Identifying how community guidelines, automated moderation rules, algorithmic biases, or user folk theories reflect underlying cultural assumptions about "acceptable" discourse.[1]
Meaning	The polysemic nature of technology; the recognition that artifacts and practices possess multiple, shifting meanings depending on the specific demographic population and their unique circumstances.	Interpreting how a digital practice (e.g., accusations of "digital blackface" or algorithmic resistance) generates unique cultural significance and solidarity within marginalized counterpublics.[1]

In applying CTDA, the objective is to reject technological determinism. The framework posits that technologies are shaped by the sociocultural contexts of their design and use, while society simultaneously organizes itself through the artifacts, ideologies, and discourses of ICT-based technoculture.[1] Therefore, any analytical instrument deployed against a platform must treat the platform's user interface not as a passive canvas, but as an active participant in the discourse.

Reddit's Technocultural Terrain and Architectural Affordances

Reddit presents a unique architectural and cultural ecosystem that serves as an ideal subject for CTDA. Unlike platforms centered on individual user profiles and ego-networks (such as Twitter or Instagram), Reddit is fundamentally structured around decentralized, topic-based communities known as subreddits.[12] This specific architectural choice profoundly influences the technocultural discourse, necessitating a precise mapping of Brock's four dimensions to the platform's digital affordances.[2]

When configuring the data scraping and sampling strategy, the LLM must be explicitly programmed to recognize the structural realities of Reddit. The platform's **Form** is defined by its pseudonymous user base, deeply nested and threaded comment structures, and the heavily influential, user-driven upvote/downvote mechanism.[13] This voting system is not merely a functional or neutral sorting tool; it is a highly charged material artifact that dictates algorithmic visibility, effectively crowdsourcing the marginalization or amplification of specific

epistemologies and enforcing a majoritarian consensus that often suppresses subaltern voices.
The **Function** on Reddit manifests in highly specific, community-bound practices. Each subreddit operates as a distinct enclave or counterpublic with its own vernacular, meme formats, inside jokes, and rhetorical structures.[14] For example, the performative practices within r/BlackPeopleTwitter operate vastly differently than those in mainstream news subreddits. To capture this, data collection tools must extract not only the original, top-level posts but also the deeply nested comment trees where these functional practices are negotiated and contested in real-time.
Belief is explicitly codified on Reddit through subreddit-specific rules, auto-moderator configurations, and the subjective enforcement of these boundaries by volunteer human moderators.[14] These moderation practices often reflect the ideological boundaries of the community, dictating who belongs and what discourse is permissible. Conversely, Reddit's historical overarching corporate "hands-off" moderation approach enabled the proliferation of toxic subcultures, such as Gamergate, demonstrating how the platform's foundational beliefs dictate cultural outcomes.[13] Analyzing moderation actions, locked threads, deleted comments (where accessible), and stickied meta-threads is crucial for uncovering the underlying beliefs of any specific enclave.[13]
Finally, **Meaning** is extracted by analyzing how specific subreddits construct shared identities and navigate algorithmic constraints. Research into digital Blackness on Reddit, for example, requires evaluating how Black users navigate the overarching, predominantly white normative culture of the platform to carve out spaces of digital solidarity, joy, and resistance.[3]
Furthermore, users actively develop folk theories regarding how algorithms interact with their identities, resulting in "algorithmic dissonance" where personal experiences clash with systemic platform biases.[11] The sampling strategy must purposively target these specific enclaves and interactions rather than relying on the platform's default "front page," which merely reflects the hegemonic center.[1]

Open WebUI: Infrastructure and Temporary Interface Deployment

To execute this highly specialized analysis without exposing sensitive cultural data or proprietary analytical prompts to commercial API providers, a heavily controlled, localized computational environment is required. Open WebUI functions as an extensible, protocol-first platform that operates entirely offline, making it the premier temporary interface for processing sensitive technocultural data.[5]
Open WebUI's architecture separates core capabilities into distinct extensibility vectors, which must be understood to properly configure the CTDA instrument [18]:

Extensibility Vector	Definition and Scope in Open WebUI	Application to CTDA Research
Tools	Scripts that extend the abilities of the LLM, allowing the model	The Python PRAW script used by the model to autonomously

	to perform custom tasks or retrieve live external data during a chat session.	scrape Reddit data in real-time.[18]
Functions	Scripts that extend the capabilities of the Open WebUI platform itself, enabling interface customization or new administrative features.	Used to build custom UI buttons for researchers to easily tag discourse or save API keys via Valves.[19]
Pipelines	Advanced workflows that offload heavy processing to external servers, transforming Open WebUI features into API-compatible workflows.	Bypassed for this specific implementation to maintain a strictly localized, offline processing environment.[19]
Skills	Reusable, markdown-based instruction sets that teach a model *how* to approach a task or execute a specific methodology.	Defining the purposive sampling protocols and CTDA analytical guidelines without bloating the core system prompt.[22]

Docker Deployment Specifications

The deployment of Open WebUI as a temporary research interface is optimally achieved via Docker. This containerized approach ensures environmental consistency, simplifies dependency management, and allows the entire analytical apparatus to be spun up and torn down rapidly without leaving residual configurations on the host machine.[5]
The deployment utilizes the GitHub Container Registry to pull the latest image. For environments requiring heavy data processing, document ingestion, and local LLM execution (such as running Ollama locally alongside the WebUI), enabling Nvidia GPU support is critical to prevent unacceptable latency during the high-volume text analysis required by CTDA.[5]
The foundational command to establish the temporary interface with CUDA acceleration is executed as follows:

Bash

```
docker run -d -p 3000:8080 --gpus all \
--add-host=host.docker.internal:host-gateway \
-v open-webui-ctda-data:/app/backend/data \
--name ctda-webui-interface --restart always \
ghcr.io/open-webui/open-webui:cuda
```

This configuration maps the internal Docker port 8080 to the host port 3000, allowing researcher access to the WebUI via localhost:3000.[5] The persistent volume mapping

(open-webui-ctda-data) is a critical component; it ensures that the SQLite database (webui.db), which stores the Modelfiles, Knowledge collections, encrypted API keys (Valves), and custom Tools, remains completely intact across container restarts or system reboots.[17]
To ensure the temporary interface remains stable during massive data ingestion, administrators should utilize basic health monitoring. Uptime Kuma can be configured to ping the publicly accessible /health endpoint, which returns a 200 OK HTTP status code when the Open WebUI service is running correctly, ensuring the system does not silently fail during prolonged scraping operations.[25]

Modelfile Configuration and System Parameters

Once the Open WebUI interface is active, the next critical phase is constructing the custom Modelfile. While Open WebUI seamlessly interfaces with backends like Ollama that utilize their own Modelfile formats, Open WebUI's internal Preset System is vastly superior for complex research. It allows administrators to "wrap" the base model (whether a local Llama 3 instance or a securely proxied API model), permanently binding specific System Prompts, Knowledge Collections, Tools, and dynamic variables directly to the agent persona.[24]
For discourse analysis, context length and deterministic generation are paramount. CTDA requires evaluating long, heavily nested Reddit comment trees to accurately capture the full semiotic exchange and map the "Function" and "Belief" dimensions. Therefore, the base model selected must support an extended context window, and the Modelfile must explicitly declare this parameter to override default limitations.[27]

Generation Parameters for Qualitative Fidelity

The Modelfile must strictly control the model's generation parameters to prioritize analytical fidelity over creative text generation. High temperatures lead to hallucinatory analysis, which fundamentally destroys the validity of any critical theoretical framework by allowing the model to invent cultural context that does not exist in the scraped text.[28]

Parameter	**Recommended Value**	**Analytical Justification for CTDA Execution**
num_ctx	32768 (or maximum supported)	Allows ingestion of entire Reddit comment trees in a single pass, preserving the relational context between users necessary for extracting functional practices.[27]
temperature	0.1	Enforces highly deterministic outputs. This ensures the model adheres strictly to the provided text and theoretical matrix rather than creatively

		extrapolating cultural contexts.[28]
top_p	0.2	Restricts the sampling pool to only the most probable tokens, further anchoring the model's analytical output to the provided CTDA theory and scraped data.[28]
repeat_penalty	1.15	Prevents the model from entering repetitive analytical loops when processing highly repetitive Reddit meme structures or copy-paste text blocks ("copypasta") common in digital discourse.[27]

The System Prompt Architecture

The System Prompt is the core directive that forces the LLM to adopt the persona of a critical digital sociologist. Standard LLM system prompts default to a "helpful, harmless, and honest" AI assistant persona, which is entirely counterproductive for critical sociological research. The prompt must explicitly instruct the model to abandon normative, deficit-based analyses and mandate the strict use of Brock's four dimensions.[1]

The system prompt within the Open WebUI Modelfile configuration must be structured as follows:

You are an expert sociotechnical researcher specializing in Critical Technocultural Discourse Analysis (CTDA) as defined by André Brock. Your sole objective is to analyze digital phenomena by integrating an analysis of technological artifacts and user discourse, strictly framed by critical cultural theory.

You must actively reject deficit-based models and normative white, Western, patriarchal hegemony. You must prioritize the epistemological standpoints of marginalized and underrepresented groups in all analyses.

When provided with scraped Reddit data, you must evaluate the text strictly across the following four dimensions:

1. FORM: Analyze the material constraints, UI elements (upvotes, threaded architecture), and algorithmic mechanisms influencing the discourse.
2. FUNCTION: Analyze the specific cultural practices, interactions, and vernacular deployed by the users within the enclave.
3. BELIEF: Identify the cultural ideologies, moderation rules, and systemic assumptions embedded in the exchange.
4. MEANING: Interpret the polysemic, shifting significance of these practices for the specific demographic population involved, mapping their digital solidarity and resistance.

Do not provide generic summaries. Ground all observations explicitly in the exact text provided,

connecting the discourse to the underlying libidinal economy and systemic structures.
By locking down this system prompt into the Modelfile template and enforcing context-to-prompt ordering, the researcher ensures the model remains fundamentally anchored to the CTDA methodology throughout the duration of the research session.[28]

Knowledge Collection and the "One Scale" System Prompt Reduction

In Open WebUI, the "Knowledge" feature provides robust Retrieval-Augmented Generation (RAG) capabilities. This allows the model to dynamically query internal databases of pre-processed literature, codebooks, or historical data to inform its real-time analysis.[22] An administrator can navigate to the "Knowledge" tab, create a new private database, and upload critical theoretical texts, such as Brock's foundational papers, literature on the libidinal economy, and prior studies concerning Reddit's moderation practices.[9]
However, a significant architectural challenge in using standard RAG for complex qualitative analysis is prompt bloat and instruction conflict. When Open WebUI retrieves chunks of text from a knowledge base via vector search, the default internal system prompt wrapped around that retrieved text often contains verbose, generic instructions (e.g., "Use the following context to answer the user's question comprehensively"). This generic instruction encourages the model to summarize the retrieved text rather than use it as a critical analytical lens, causing the LLM to lose focus on the primary CTDA framework established in the Modelfile.
To resolve this, the system requires reducing the system prompt recommendations for the knowledge collection to a singular, standardized scale—referred to as the "One Scale" optimization.[4] This directive means engineering a highly constrained, unified qualitative evaluation matrix that the model must strictly adhere to when processing retrieved knowledge. Instead of allowing the RAG system to generate open-ended narrative comparisons, the knowledge retrieval process is bottlenecked into evaluating the presence and intensity of CTDA variables on a singular binary or Likert-style scale.

Implementing the "One Scale" RAG Matrix

The standard RAG template in Open WebUI's advanced settings must be completely rewritten. The reduced, "One Scale" prompt template strips away extraneous conversational instructions and forces the model to evaluate all retrieved context through the CTDA matrix before injecting it into the main conversation flow.[4]
The "One Scale" RAG configuration is defined as follows:
CONTEXT RETRIEVED:
{context}
TASK INSTRUCTION:
Evaluate the retrieved context strictly on the CTDA One-Scale matrix. For each of the four CTDA dimensions, assign a binary relevance score (0=Absent/Irrelevant to current data, 1=Present/Highly Relevant) and provide a single-sentence critical justification linking the context to the active Reddit data. Do not summarize the context. Do not extrapolate beyond

the text.
REQUIRED OUTPUT FORMAT:

- Form: [0/1] - [Critical Justification]
- Function: [0/1] - [Critical Justification]
- Belief: [0/1] - [Critical Justification]
- Meaning: [0/1] - [Critical Justification]

By enforcing this reduced, mathematically structured output, the agent processes all stored knowledge collections—whether they are prior academic papers on digital blackness, documentation on Reddit's API, or historical scraped datasets—through a uniform analytical lens.[32] This directly prevents the model from suffering "contextual drift." It ensures that every piece of injected theoretical knowledge is immediately quantified and qualified exactly according to Brock's framework, transforming the LLM from a generic conversationalist into a precision sociological instrument.[2]
When Native Function Calling (Agentic Mode) is enabled, frontier models can autonomously explore these Knowledge Bases using built-in tools like query_knowledge_bases and search_knowledge_files, retrieving Brock's texts and instantly converting them into the "One Scale" matrix to evaluate live Reddit data.[22]

Collection Skills: Engineering the Purposive Sampling Protocol

While Knowledge provides the theoretical data, Open WebUI's distinct "Skills" feature dictates the methodological execution. Skills in Open WebUI are defined as reusable, markdown-based instruction sets that teach a model *how* to approach a task or execute a workflow, rather than being executable Python scripts or static data repositories.[22]
For CTDA, relying on an LLM to automatically know how to sample Reddit data is a critical error. Left to its own devices, an LLM will default to convenience sampling, targeting mainstream subreddits (e.g., r/AskReddit, r/news) that center white, normative discourse, thereby violating the core tenets of the framework.[1] To prevent this, researchers must configure a specific "Collection Skill" that dictates the purposive sampling methodology.

Configuring the CTDA Collection Skill

To create this skill, the administrator navigates to **Workspace → Skills** and creates a new entry named "CTDA Purposive Sampling Protocol".[22] The content of this skill is written in markdown and serves as a highly specific algorithmic playbook for the model.
The Collection Skill markdown is configured as follows:

CTDA Purposive Sampling Protocol

When tasked with collecting or analyzing Reddit data, you must strictly adhere to this purposive sampling methodology to avoid the whitewashing of generality.

Step 1: Enclave Identification

Do not sample from default or mainstream subreddits unless explicitly instructed to analyze hegemonic behavior. You must actively seek out and identify subreddits that function as counterpublics or marginalized enclaves (e.g., subreddits centered on Black digital networks, LGBTQ+ communities, or specific minority intersections).

Step 2: Data Extraction parameters

When invoking the Reddit scraper tool, you must request deep comment threads (depth > 10). Surface-level posts only represent the "Form" and initial "Belief" of a community; the nested discourse is required to evaluate the "Function" and the shifting "Meaning" of the community interactions.

Step 3: Artifact Recognition

Actively scan the scraped metadata for indicators of systemic friction or algorithmic dissonance. Pay immediate attention to:

- Threads with high engagement but low upvote ratios (indicating brigading or outside interference).
- Locked threads (indicating a clash between community 'Beliefs' and platform moderation).
- Stickied moderator comments (representing explicit codification of enclave rules).

In Open WebUI, Skills are accessed dynamically. A user can inject this skill into a chat by typing the $ symbol followed by the skill name, which injects the full markdown content directly into the system prompt for that specific conversation.[22] Alternatively, this skill can be permanently attached to the CTDA Modelfile using Open WebUI's lazy-loading architecture. In this mode, the model reads the skill's description and autonomously calls a view_skill builtin tool whenever it determines it needs methodological guidance on how to sample data.[22] This preserves valuable token space in the context window while ensuring the purposive sampling protocol is always strictly enforced.

The Python Toolset: Engineering the PRAW Scraper

With the Modelfile configured, the Knowledge Base optimized to the "One Scale", and the Collection Skills defining the sampling methodology, the agent requires a mechanical mechanism to interact with the external world. Open WebUI supports the integration of custom "Tools"—pure Python classes that extend the LLM's capabilities, allowing it to perform custom tasks like real-time data extraction and execute them locally in the environment.[18]

To scrape Reddit effectively for CTDA, the tool must leverage PRAW (Python Reddit API Wrapper). PRAW provides direct, programmatic access to Reddit's official API. Using an official wrapper ensures stable, compliant data extraction, bypassing the fragile nature of raw HTML scraping (like BeautifulSoup), which breaks whenever Reddit updates its DOM.[6] PRAW also automatically handles pagination and the complex resolution of "load more comments" forests,

ensuring the data structure remains intact.[6]

Security and Asynchronous Execution

Embedding hardcoded API credentials within a Python script is a severe security vulnerability. Open WebUI resolves this through "Valves," a Pydantic-based configuration system. Valves allow administrators to securely input API keys via the WebUI settings interface, storing them encrypted in the SQLite database and exposing them dynamically to the Python Tool class at runtime.[20]

Furthermore, the Python class must conform to strict structural requirements. It requires an __init__ method, comprehensive docstrings to inform the LLM of the tool's specific purpose, and crucially, asynchronous execution.[20] If the PRAW library makes synchronous HTTP requests on the main thread, it will block Open WebUI's WebSocket communication, causing the UI to freeze and crash.[5] The tool must wrap PRAW's synchronous calls in an asyncio event loop executor.

CTDA-Aligned PRAW Implementation

A standard Reddit scraper simply pulls the post title, body text, and author.[36] This is insufficient for Brock's framework. A CTDA-compliant Python tool must preemptively extract and map data directly to the framework's dimensions. It must extract the "Form" (upvote ratios, scores, flair, stickied status), the "Function" (the deeply nested comment trees where discourse happens), and the "Belief" (moderator interventions).[2]

The following Python code represents the complete, production-ready Open WebUI Tool designed specifically for this task:

Python

```
"""
title: CTDA Reddit Discourse Scraper
author: CTDA System Architect
description: Extracts Reddit posts and nested comment trees via PRAW, structuring metadata directly for Critical Technocultural Discourse Analysis (Form, Function, Belief, Meaning).
"""

import os
import asyncio
from typing import List, Dict, Any, Optional
from pydantic import BaseModel, Field
import praw

class Tools:
    class Valves(BaseModel):
```

```
    REDDIT_CLIENT_ID: str = Field(
        default="",
        description="The Client ID from Reddit App Preferences."
    )
    REDDIT_CLIENT_SECRET: str = Field(
        default="",
        description="The Client Secret from Reddit App Preferences."
    )
    REDDIT_USER_AGENT: str = Field(
        default="CTDA_Research_Bot:v1.0 (by /u/researcher)",
        description="A unique, descriptive User-Agent string."
    )

def __init__(self):
    """
    Initializes the Tool and sets up the PRAW client placeholder.
    The PRAW client is instantiated dynamically in the async methods
    to ensure the Valves are loaded securely from the Open WebUI database.
    """
    self.valves = self.Valves()

def _get_reddit_client(self) -> praw.Reddit:
    """Helper method to initialize the PRAW client using secure Valves."""
    if not self.valves.REDDIT_CLIENT_ID or not self.valves.REDDIT_CLIENT_SECRET:
        raise ValueError("Reddit API credentials are not configured in the Tool Valves.")

    return praw.Reddit(
        client_id=self.valves.REDDIT_CLIENT_ID,
        client_secret=self.valves.REDDIT_CLIENT_SECRET,
        user_agent=self.valves.REDDIT_USER_AGENT
    )

async def scrape_subreddit_discourse(
    self,
    subreddit_name: str,
    sort_by: str = "hot",
    limit: int = 5,
    comment_depth: int = 20
) -> Dict[str, Any]:
    """
    Scrapes a specified subreddit to extract posts and nested comments for CTDA analysis.

    :param subreddit_name: The name of the subreddit without 'r/' (e.g., 'BlackPeopleTwitter').
```

```
    :param sort_by: Sorting method: 'hot', 'new', 'top', or 'controversial'.
    :param limit: Number of top-level submissions to retrieve.
    :param comment_depth: Maximum number of comments to extract per submission to
capture discourse function.
    :return: A JSON-serializable dictionary mapping data to CTDA dimensions.
    """
    try:
        # Execute synchronous PRAW network calls within an async executor
        # to avoid blocking the Open WebUI main thread and WebSockets.
        loop = asyncio.get_event_loop()
        result = await loop.run_in_executor(
            None,
            self._sync_scrape,
            subreddit_name,
            sort_by,
            limit,
            comment_depth
        )
        return result
    except Exception as e:
        return {"error": f"Failed to execute CTDA scrape on {subreddit_name}: {str(e)}"}

def _sync_scrape(
    self,
    subreddit_name: str,
    sort_by: str,
    limit: int,
    comment_depth: int
) -> Dict[str, Any]:
    """Synchronous backend logic for PRAW execution and CTDA mapping."""
    reddit = self._get_reddit_client()
    subreddit = reddit.subreddit(subreddit_name)

    if sort_by == "new":
        submissions = subreddit.new(limit=limit)
    elif sort_by == "top":
        submissions = subreddit.top(limit=limit)
    elif sort_by == "controversial":
        submissions = subreddit.controversial(limit=limit)
    else:
        submissions = subreddit.hot(limit=limit)

    discourse_data = {
```

```
    "metadata": {
        "target_subreddit": subreddit_name,
        "sort_method": sort_by,
        "analytical_framework": "Critical Technocultural Discourse Analysis (Brock, 2018)"
    },
    "submissions":
}

for sub in submissions:
    # Capture structural elements mapping to CTDA "Form"
    sub_data = {
        "form_elements": {
            "submission_id": sub.id,
            "score": sub.score,
            "upvote_ratio": sub.upvote_ratio,
            "author_flair": sub.author_flair_text,
            "is_locked": sub.locked,
            "is_stickied": sub.stickied
        },
        "discourse_content": {
            "title": sub.title,
            "selftext": sub.selftext,
            "author": str(sub.author) if sub.author else "[deleted]"
        },
        "nested_discourse":
    }

    # Extract comment trees mapping to CTDA "Function" and "Belief"
    sub.comments.replace_more(limit=0) # Flatten tree, discard 'load more' stubs
    comment_count = 0

    for comment in sub.comments.list():
        if comment_count >= comment_depth:
            break

        comment_data = {
            "form_elements": {
                "score": comment.score,
                "is_submitter": comment.is_submitter,
                "is_stickied": comment.stickied # Highlights Moderator Beliefs
            },
            "author": str(comment.author) if comment.author else "[deleted]",
            "body": comment.body
```

```
            }
            sub_data["nested_discourse"].append(comment_data)
            comment_count += 1

        discourse_data["submissions"].append(sub_data)

    return discourse_data
```

This specific implementation preemptively structures the JSON payload to mirror the CTDA dimensions. By explicitly defining keys such as form_elements (containing the upvote ratios and lock status) and separating them from the nested_discourse (containing the linguistic interaction), the Python script forces the LLM to ingest the data already categorized by the framework's variables. This architectural choice drastically reduces hallucinations, ensuring the agent constantly recognizes the material constraints of the platform alongside the raw textual content.[2]

Pipeline Synthesis: Execution of the Analytical Instrument

With the Docker interface running locally, the Modelfile locked down with the appropriate parameters, the Knowledge Base optimized to the "One Scale", the purposive sampling Collection Skill configured, and the Python PRAW tool active, the system is fully instrumented. The execution of a CTDA research session follows a strict, autonomous pipeline initiated by the researcher.

In the Open WebUI chat interface, the researcher calls upon the $CTDA Purposive Sampling Protocol skill and submits a prompt to the agent: *"Execute a CTDA on the intersection of moderation practices and user vernacular regarding algorithmic dissonance. Sample the top 10 controversial posts from a relevant enclave."*

Because the Collection Skill dictates the sampling methodology, the model autonomously selects an appropriate subreddit—such as r/BlackPeopleTwitter or a similar counterpublic—rather than defaulting to a normative sub.[15] The LLM outputs a specialized JSON request invoking the scrape_subreddit_discourse tool. The Open WebUI backend instantiates the Tools class, injects the secure API keys via the Valves, and executes the PRAW script asynchronously in the background.[5]

Simultaneously, the model queries the configured Knowledge Base.[31] Because the RAG system prompt was reduced to the "One Scale" evaluation matrix, the vector database returns contextual chunks from André Brock's literature, pre-formatted as highly constrained binary evaluations of Form, Function, Belief, and Meaning.[4] The model ingests these theoretical anchors, utilizing them to ground its analysis.

In the final phase, the LLM synthesizes the inputs. The agent processes the form_elements (e.g., heavily downvoted comments or locked threads indicative of algorithmic brigade attacks) and correlates them with the nested_discourse (the Function). Through the strict lens of the

"One Scale" RAG data, it interprets how the subreddit's moderation rules (Belief) are actively deployed to protect the enclave from outside interference, ultimately generating a profound understanding of the community's digital solidarity and resistance (Meaning).[1]
Because the agent is governed by the strict Modelfile temperature parameters (0.1) and the overarching system prompt rejecting deficit models, the final output avoids the pathological framing ubiquitous in legacy digital sociology. Instead, it produces a deep, critical reading of the libidinal economy driving the subreddit's technoculture, successfully operating as an automated, highly precise instrument for Critical Technocultural Discourse Analysis.

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